

# Anemia, CBC Interpretation, and Transfusion Foundations

## Part VII. Hematology and Oncology

This chapter is an original Medvale framework chapter built to turn anemia into a usable clinical reasoning system. The uploaded reference begins with anemia basics, reticulocyte response, transfusion pearls, iron deficiency anemia, thalassemias, sideroblastic anemia, anemia of chronic disease, aplastic anemia, vitamin B12 deficiency, folate deficiency, and the opening framework for hemolytic anemia. This chapter expands those topics into a paragraph-rich, graph-RAG friendly study resource with diagnostic links, numerical anchors, management thresholds, and integrated cases.

Coverage note: this is the first hematology segment. It intentionally emphasizes CBC interpretation, anemia classification, microcytic/normocytic/macrocyclic anemia, transfusion basics, and the initial hemolysis framework. Later hematology chapters can deepen hemolytic anemias, hemoglobinopathies, platelets/coagulation, malignant hematology, and oncology.

| Core task                       | What the learner should be able to do after this chapter                                                                                                                                           |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recognize anemia patterns       | Use hemoglobin/hematocrit, MCV, RDW, reticulocyte index, and smear clues to separate production failure from blood loss or hemolysis.                                                              |
| Classify anemia mechanistically | Link low reticulocyte states to marrow substrate failure, chronic inflammation, kidney disease, or marrow replacement; link high reticulocyte states to bleeding or hemolysis.                     |
| Use iron studies                | Distinguish iron deficiency, anemia of inflammation, thalassemia trait, and sideroblastic processes using ferritin, serum iron, transferrin/TIBC, transferrin saturation, RDW, and smear patterns. |
| Prevent harm                    | Know when transfusion is urgent, when iron should not be given blindly, why B12 deficiency must not be missed, and how transfusion reactions present.                                              |

# 1. Why Anemia Matters Clinically

Anemia is not a diagnosis; it is evidence that oxygen-carrying capacity has fallen or that the measured red cell concentration is diluted. A patient can have a hemoglobin of 6.5 g/dL and be walking if the decline was slow, or can have a hemoglobin of 10 g/dL and be unstable if blood loss is acute and intravascular volume is collapsing. Clinical reasoning therefore begins with three linked questions: Is the patient hemodynamically stable? Is the anemia acute or chronic? Is the marrow responding appropriately?

A red blood cell mass deficit lowers arterial oxygen content because most oxygen is carried bound to hemoglobin rather than dissolved in plasma. Compensatory physiology includes increased cardiac output, increased oxygen extraction, and increased 2,3-BPG, which shifts the oxyhemoglobin curve to the right and helps unload oxygen to tissues. These compensations take time. Acute hemorrhage may first present with tachycardia, orthostasis, syncope, or hypotension before the hemoglobin concentration has fully equilibrated downward.

Numerical anchors: adult anemia is commonly defined near hemoglobin <13 g/dL in men and <12 g/dL in nonpregnant women, but clinical action depends on symptoms, rate of change, comorbid cardiopulmonary disease, active bleeding, and the suspected cause. A restrictive transfusion strategy commonly considers red cell transfusion around hemoglobin <7 g/dL in stable hospitalized adults, while higher thresholds may be appropriate in selected patients with active ischemia, major bleeding, or limited cardiopulmonary reserve.

## Clinical priority order

- 1 First assess stability: mental status, blood pressure, orthostasis, heart rate, chest pain, dyspnea, melena/hematemesis, trauma, and ongoing blood loss.
- 2 Second confirm true anemia: compare with prior CBCs, evaluate hydration status, and consider dilutional anemia after large-volume fluids.
- 3 Third determine marrow response using reticulocyte count or corrected reticulocyte index.
- 4 Fourth classify by RBC size and smear: microcytic, normocytic, macrocytic, or mixed populations.
- 5 Fifth identify mechanism: substrate deficiency, inflammation/hepcidin biology, marrow failure, kidney disease, blood loss, or hemolysis.

A common exam trap is to chase iron studies or B12 levels before recognizing hemorrhagic shock. Another trap is to transfuse only because a hemoglobin value is low without asking whether the patient is bleeding, ischemic, symptomatic, or chronically adapted. The number matters, but the patient matters more.

# 2. CBC Interpretation: Turning Numbers Into Physiology

A CBC can be treated as a small physiologic dataset. Hemoglobin and hematocrit estimate oxygen-carrying concentration; MCV estimates average red cell size; RDW estimates variation in size; WBC and platelet counts reveal whether anemia is isolated or part of a multilineage marrow problem. The reticulocyte count is the bridge between a static CBC and marrow physiology.

| CBC variable  | High-yield interpretation                                                                                                     | Common traps                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Hemoglobin    | Most direct estimate of oxygen-carrying red cell concentration. Used for anemia severity and transfusion context.             | May lag after acute bleeding until plasma volume equilibrates; can be diluted after IV fluids.                               |
| Hematocrit    | Approximate fraction of blood volume occupied by RBCs. Roughly Hct is about 3 times Hb in many stable patients.               | Can be misleading in dehydration, massive fluid resuscitation, or acute hemorrhage.                                          |
| MCV           | Microcytic <80 fL, normocytic 80-100 fL, macrocytic >100 fL. Best first sorting variable after reticulocyte response.         | Early anemia of any cause may be normocytic; mixed microcytic and macrocytic processes can produce a deceptively normal MCV. |
| RDW           | Measures red cell size variability. Often high in iron deficiency and mixed deficiencies; often normal in thalassemia trait.  | Normal RDW does not exclude disease; interpret with smear and RBC count.                                                     |
| Platelets/WBC | Pancytopenia suggests marrow failure, marrow infiltration, severe nutritional deficiency, hypersplenism, or systemic illness. | Isolated anemia favors RBC-specific mechanisms; multilineage abnormalities demand broader workup.                            |
| Reticulocytes | Shows marrow response. High suggests blood loss or hemolysis; low/inappropriately normal suggests underproduction.            | Raw retic percent can look normal or high in severe anemia; correct it for the degree of anemia.                             |

## Reticulocyte index

Reticulocytes are young RBCs released from marrow. In anemia, a healthy marrow should increase production. The corrected reticulocyte index adjusts for the fact that a reticulocyte percentage is relative to a reduced RBC denominator. A low or inappropriately normal reticulocyte index means the marrow is not keeping up; a high index means the marrow senses anemia and is responding, usually because RBCs are being lost or destroyed.

Practical interpretation: reticulocyte index  $>2$  suggests an appropriate marrow response; reticulocyte index  $<2$  suggests inadequate production. Very severe hemolysis or brisk hemorrhage can produce much higher reticulocyte responses if marrow substrates and erythropoietin signaling are intact.

## Peripheral smear as a diagnostic amplifier

The smear can convert a broad differential into a pattern diagnosis. Microcytosis with pencil cells supports iron deficiency; target cells suggest thalassemia, liver disease, or hemoglobinopathy; hypersegmented neutrophils support megaloblastic anemia; schistocytes suggest microangiopathic hemolysis; spherocytes suggest hereditary spherocytosis or autoimmune hemolysis; bite cells suggest oxidant injury such as G6PD deficiency; teardrop cells suggest marrow infiltration or fibrosis.

## Original diagnostic map: anemia from CBC to mechanism

|                                       |                                       |                                                  |                            |                        |
|---------------------------------------|---------------------------------------|--------------------------------------------------|----------------------------|------------------------|
| Low Hb / Hct confirmed                | Assess stability and bleeding         | Reticulocyte index                               | $>2$ : loss or destruction | $<2$ : underproduction |
| If unstable: resuscitate, type/source | Hemolysis labs: LDH, haptoglobin, DAT | MCV + smear + iron/B12/folate/renal/inflammation |                            |                        |

Figure 7A-1. A screen-safe, text-wrapped anemia flow map. The first decision is not microcytic versus macrocytic; the first decision is stability. Once immediate bleeding risk is addressed, the reticulocyte index separates marrow underproduction from RBC loss/destruction.

## 3. Transfusion Foundations and Reactions

Red cell transfusion is a treatment for inadequate oxygen delivery, not a treatment for every low hemoglobin value. The decision integrates hemoglobin, symptoms, rate of decline, active bleeding, coronary or pulmonary disease, and the likelihood that the underlying process can be treated quickly. In stable patients, one unit of packed RBCs often raises hemoglobin by about 1 g/dL and hematocrit by about 3 percentage points. Reassess after each unit rather than ordering reflexive multiunit transfusions.

A modern restrictive strategy commonly uses a hemoglobin threshold near 7 g/dL for stable hospitalized adults, but thresholds are not a substitute for bedside judgment. Active hemorrhage, hemodynamic instability, symptomatic anemia, acute coronary syndrome, major surgery, or chronic transfusion programs may require individualized targets.

| Product             | Main contents                                                             | Typical clinical use                                                                                                                                             | Key cautions                                                                                                                       |
|---------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Packed RBCs         | Red cells with most plasma removed.                                       | Symptomatic anemia, active blood loss, perioperative anemia, severe chronic anemia when other therapy is too slow.                                               | ABO compatibility; risk of hemolytic reaction, TACO, TRALI, infection, alloimmunization, iron overload with repeated transfusions. |
| Platelets           | Platelet concentrates from donor pool or apheresis.                       | Bleeding with thrombocytopenia or platelet dysfunction; procedural prophylaxis depending on platelet count and site.                                             | Less useful in immune platelet destruction unless life-threatening bleeding; may worsen thrombosis risk in some microangiopathies. |
| Fresh frozen plasma | Coagulation factors including fibrinogen, factors II, V, VII, IX, X, XI.  | Bleeding with multiple factor deficiencies, massive transfusion, urgent warfarin reversal when PCC not available, liver disease with bleeding/procedure context. | Volume load; not for isolated mild INR elevation without bleeding risk.                                                            |
| Cryoprecipitate     | Fibrinogen, factor VIII, factor XIII, von Willebrand factor, fibronectin. | Hypofibrinogenemia in bleeding, DIC, massive transfusion, obstetric hemorrhage.                                                                                  | Does not replace all coagulation factors; use fibrinogen level and bleeding context.                                               |

## Transfusion reaction framework

Any fever, chills, dyspnea, hypotension, back pain, flank pain, chest pain, hemoglobinuria, or unexplained shock during transfusion should be treated as a transfusion reaction until proven otherwise. Stop the transfusion, maintain IV access with normal saline, verify patient/product identification, notify the blood bank, and send the appropriate post-reaction workup.

| Reaction              | Timing and clues                                                                                               | Mechanism                                                                           | Immediate response                                                                               |
|-----------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Acute hemolytic       | Minutes to hours: fever, chills, flank/back pain, hypotension, hemoglobinuria, DIC, renal failure.             | Usually ABO incompatibility with intravascular complement-mediated RBC destruction. | Stop transfusion; aggressive supportive care; fluids; vasopressors if needed; blood bank workup. |
| Delayed hemolytic     | Days to weeks: falling Hb, jaundice, mild fever, positive DAT, new alloantibody.                               | Anamnestic antibody response to prior RBC antigen exposure.                         | Supportive care; avoid implicated antigen in future transfusions.                                |
| Febrile nonhemolytic  | Fever/chills without hemolysis, often during or shortly after transfusion.                                     | Recipient cytokine response or antibodies to donor leukocytes.                      | Stop and evaluate; antipyretics after hemolysis excluded; leukoreduced products reduce risk.     |
| Allergic/anaphylactic | Urticaria to wheezing, hypotension, anaphylaxis.                                                               | Reaction to donor plasma proteins; severe reactions in IgA deficiency can occur.    | Antihistamines for mild; epinephrine and emergency care for anaphylaxis.                         |
| TACO                  | Dyspnea, hypertension, pulmonary edema, positive fluid balance, elevated BNP; often older/heart/renal disease. | Circulatory volume overload.                                                        | Stop/slow transfusion, oxygen, diuretics; prevent with slower rate and diuretic strategy.        |
| TRALI                 | Acute hypoxemia and noncardiogenic pulmonary edema within 6 hours, often hypotension/fever.                    | Donor anti-leukocyte antibodies or inflammatory priming causing lung injury.        | Supportive oxygen/ventilation; avoid diuretics unless volume overload also present.              |

## 4. General Diagnostic Approach to Anemia

After stabilization, anemia is most efficiently approached as a two-axis problem. The first axis is marrow response: high reticulocyte index means the marrow is responding to loss/destruction; low reticulocyte index means the marrow cannot produce enough RBCs. The second axis is RBC morphology: microcytic anemia points to hemoglobin synthesis problems, macrocytic anemia points to DNA synthesis or membrane/liver/reticulocyte effects, and normocytic anemia may be early disease, kidney disease, inflammation, marrow failure, or acute blood loss.

| Pattern               | Most important mechanisms                                                                                                                          | First tests                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Microcytic, low retic | Iron deficiency, anemia of inflammation, thalassemia trait, sideroblastic anemia, lead toxicity.                                                   | Ferritin, iron, TIBC/transferrin, transferrin saturation, RDW, smear, sometimes hemoglobin electrophoresis.      |
| Normocytic, low retic | Anemia of chronic disease, CKD/low EPO, endocrine disease, marrow failure, marrow infiltration, aplastic anemia, early iron/B12/folate deficiency. | Creatinine/eGFR, inflammatory context, retic index, smear, WBC/platelets, iron/B12/folate, TSH when appropriate. |
| Macrocytic, low retic | B12 deficiency, folate deficiency, medications, alcohol, liver disease, hypothyroidism, myelodysplasia.                                            | B12, folate, methylmalonic acid, homocysteine, smear, LFTs, TSH, medication/alcohol review.                      |
| Any MCV, high retic   | Bleeding or hemolysis; MCV may rise from reticulocytosis because reticulocytes are larger than mature RBCs.                                        | Bleeding evaluation, LDH, haptoglobin, indirect bilirubin, DAT, smear, urine hemoglobin.                         |

Mixed anemia is common. A patient with chronic kidney disease may also have iron deficiency from GI bleeding; a patient with alcohol use may have folate deficiency and marrow suppression; a patient with inflammatory bowel disease may have anemia of inflammation plus iron deficiency. A normal MCV does not guarantee a single process.

## 5. Microcytic Anemia: Hemoglobin Synthesis Problems

Microcytosis means RBCs are small, usually because hemoglobin synthesis is impaired. Hemoglobin production can fail because iron is absent, iron is trapped away from marrow by inflammation, globin chain synthesis is genetically reduced, or heme synthesis is abnormal. The most useful sorting tests are ferritin, serum iron, TIBC/transferrin, transferrin saturation, RDW, RBC count, smear, and clinical context.

| Diagnosis              | Ferritin                           | Serum iron | TIBC/transferrin | RDW                   | Smear/clinical clue                                                                                 |
|------------------------|------------------------------------|------------|------------------|-----------------------|-----------------------------------------------------------------------------------------------------|
| Iron deficiency        | Low (unless inflammation masks it) | Low        | High             | High                  | Pencil cells, hypochromia, pica, koilonychia; in men/postmenopausal women search for GI blood loss. |
| Anemia of inflammation | Normal/high                        | Low        | Low or normal    | Normal or mildly high | Chronic infection, autoimmune disease, malignancy, CKD; hepcidin traps iron in macrophages.         |

| Diagnosis            | Ferritin    | Serum iron  | TIBC/transferrin | RDW          | Smear/clinical clue                                                                             |
|----------------------|-------------|-------------|------------------|--------------|-------------------------------------------------------------------------------------------------|
| Thalassemia trait    | Normal/high | Normal/high | Normal           | Often normal | Target cells, high/normal RBC count despite low MCV; electrophoresis helps beta-thalassemia.    |
| Sideroblastic anemia | High        | High        | Normal/low       | Often high   | Ring sideroblasts in marrow; alcohol, lead, isoniazid, chloramphenicol, copper deficiency, MDS. |

## Iron deficiency anemia

Iron deficiency is common, but its cause matters more than the label. In menstruating patients it often reflects menstrual blood loss or pregnancy-related demand; in adult men and postmenopausal women it should trigger evaluation for occult GI blood loss unless another clear explanation exists. Low ferritin is highly specific for depleted iron stores, but ferritin is also an acute-phase reactant and may be misleadingly normal or high when inflammation, liver disease, infection, or malignancy coexist.

Iron deficiency moves through stages: iron stores fall first, then transferrin saturation falls, then hemoglobin synthesis becomes impaired, then microcytosis and hypochromia become obvious. Symptoms can include fatigue, exertional dyspnea, palpitations, pica, pagophagia, restless legs, glossitis, brittle nails, and poor exercise tolerance.

Treatment is iron replacement plus source control. Oral iron is appropriate for many stable patients, but alternate-day dosing can improve absorption and tolerability in some patients because hepcidin rises after iron ingestion. IV iron is useful when oral iron fails, malabsorption exists, ongoing losses exceed oral replacement, rapid repletion is needed, inflammatory bowel disease limits tolerance, or heart failure/CKD protocols apply. Transfusion is reserved for severe, symptomatic, unstable, or actively bleeding patients rather than simple iron depletion.

## Anemia of inflammation/chronic disease

Inflammation increases hepcidin, a liver-derived hormone that degrades ferroportin and blocks iron export from enterocytes and macrophages. The marrow is iron-restricted even though total body iron may be normal or high. This is why serum iron is low but ferritin is normal or high and TIBC is low or normal. The anemia is often normocytic early and microcytic later. Treatment is directed at the underlying inflammatory, infectious, malignant, renal, or autoimmune condition; empiric iron alone may not help unless true iron deficiency coexists.

## Thalassemias

Thalassemias are inherited disorders of globin chain production. In beta-thalassemia, beta-chain synthesis is reduced, leaving excess alpha chains that damage RBC precursors and circulating RBCs. In alpha-thalassemia, alpha-chain deletion severity ranges from silent carrier states to hemoglobin H disease and hydrops fetalis. Thalassemia trait can look like iron deficiency because both are microcytic, but thalassemia often has a disproportionately low MCV with a relatively preserved or high RBC count and normal RDW.

| Thalassemia state       | Genetic/physiologic idea             | Clinical pattern                                                                                                                       | Diagnostic clue                                                                |
|-------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Beta-thalassemia minor  | One beta-globin allele reduced.      | Mild microcytic anemia; usually no transfusion requirement.                                                                            | Hemoglobin electrophoresis often shows increased HbA2 and sometimes HbF.       |
| Beta-thalassemia major  | Severe beta-chain absence/reduction. | Severe anemia in infancy after fetal hemoglobin falls; marrow expansion, skeletal changes, hepatosplenomegaly, transfusion dependence. | High HbF, absent/reduced HbA; chronic transfusions can cause iron overload.    |
| Alpha-thalassemia trait | Two alpha-gene deletions.            | Mild microcytosis; often confused with iron deficiency.                                                                                | Electrophoresis may be normal after newborn period; genetic testing if needed. |
| Hemoglobin H disease    | Three alpha-gene deletions.          | Moderate to severe hemolytic anemia; beta-chain tetramers form HbH.                                                                    | HbH on testing; variable transfusion needs.                                    |
| Hydrops fetalis         | Four alpha-gene deletions.           | Usually fatal without fetal intervention; gamma tetramers form Hb Bart's with poor oxygen delivery.                                    | Severe fetal anemia/hydrops.                                                   |

## Sideroblastic anemia

Sideroblastic anemia is a heme synthesis problem in which iron enters erythroid precursors but cannot be incorporated normally into heme, leading to mitochondrial iron accumulation around the nucleus. The marrow finding is ring sideroblasts. Causes include myelodysplastic syndromes, alcohol, lead, isoniazid, chloramphenicol, linezolid, copper deficiency, and congenital enzymatic defects. Iron studies often show increased iron and ferritin, which distinguishes it from classic iron deficiency.

## 6. Normocytic Anemia: Normal Size Does Not Mean Normal Physiology

Normocytic anemia has a broad differential because many processes begin before RBC size changes. The reticulocyte index is especially important. A high reticulocyte index suggests acute bleeding or hemolysis. A low reticulocyte index suggests underproduction from inflammation, kidney disease, endocrine disease, marrow failure, marrow infiltration, early nutritional deficiency, or drug/toxin effects.

Kidney disease causes anemia largely through inadequate erythropoietin production, reduced RBC lifespan, inflammation, iron restriction, and blood loss from dialysis or phlebotomy. Anemia of CKD is usually normocytic and underproductive. Before erythropoiesis-stimulating agents are used, iron status should be evaluated and corrected because functional or absolute iron deficiency limits response.

## Aplastic anemia

Aplastic anemia is bone marrow failure causing pancytopenia: anemia, neutropenia, and thrombocytopenia. Symptoms therefore combine oxygen delivery problems, infection risk, and bleeding/bruising. Causes include immune-mediated idiopathic disease, medications, chemotherapy, radiation, benzene, viral infections, inherited marrow failure syndromes, and pregnancy-associated forms. The marrow is hypocellular rather than crowded with malignant cells.

| Finding            | Aplastic anemia interpretation                                                                                                                                            |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CBC                | Pancytopenia, often with normocytic anemia and low reticulocytes.                                                                                                         |
| Smear              | Few cells rather than dysplastic blasts; can be nonspecific.                                                                                                              |
| Bone marrow biopsy | Hypocellular marrow with fatty replacement and loss of hematopoietic progenitors.                                                                                         |
| Main dangers       | Severe infection from neutropenia, bleeding from thrombocytopenia, symptomatic anemia.                                                                                    |
| Treatment logic    | Remove offending cause, supportive transfusions/antimicrobials, immunosuppression in selected patients, and hematopoietic stem cell transplant in appropriate candidates. |

## 7. Macrocytic Anemia: DNA Synthesis, Membranes, and Reticulocytes

Macrocytosis is divided into megaloblastic and nonmegaloblastic patterns. Megaloblastic anemia reflects impaired DNA synthesis, classically from vitamin B12 or folate deficiency, and shows macro-ovalocytes and hypersegmented neutrophils. Nonmegaloblastic macrocytosis can result from alcohol, liver disease, hypothyroidism, reticulocytosis, medications, or myelodysplastic syndromes.

| Cause                  | Mechanism                                                                     | Key clues                                                                                                                                            | High-yield tests                                                                                                   |
|------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Vitamin B12 deficiency | Impaired DNA synthesis plus neurologic injury from myelin pathway disruption. | Glossitis, neuropathy, posterior column signs, cognitive changes, pernicious anemia, ileal disease/resection, vegan diet, metformin/PPI association. | B12 level, methylmalonic acid high, homocysteine high, intrinsic factor antibodies if pernicious anemia suspected. |
| Folate deficiency      | Impaired DNA synthesis without the same neurologic syndrome.                  | Alcohol use, poor diet, pregnancy, hemolysis, methotrexate, trimethoprim, phenytoin, malabsorption.                                                  | Folate level, homocysteine high, methylmalonic acid normal.                                                        |
| Alcohol/liver disease  | Membrane lipid changes, marrow toxicity, nutritional overlap.                 | Macrocytosis can precede anemia; target cells may occur in liver disease.                                                                            | LFTs, alcohol history, smear, folate/B12 if uncertain.                                                             |
| Reticulocytosis        | Reticulocytes are larger than mature RBCs.                                    | High retic index in bleeding or hemolysis.                                                                                                           | Retic count, hemolysis labs, bleeding evaluation.                                                                  |
| Myelodysplasia         | Clonal ineffective hematopoiesis.                                             | Older patient, cytopenias, dysplastic cells, macrocytosis not explained by vitamin deficiency.                                                       | Smear and bone marrow evaluation.                                                                                  |

## Vitamin B12 deficiency

Vitamin B12 is required for methionine synthase and methylmalonyl-CoA mutase pathways. Deficiency impairs DNA synthesis, producing megaloblastic anemia, and can damage the nervous system, producing peripheral neuropathy, loss of vibration/position sense, ataxia, spasticity, cognitive changes, and subacute combined degeneration. The neurologic disease can be prominent even when anemia is mild or absent.

B12 stores in the liver can last for years, so deficiency usually reflects prolonged malabsorption or dietary restriction. Causes include pernicious anemia from autoimmune intrinsic factor deficiency, gastrectomy, ileal disease or resection, bacterial overgrowth, fish tapeworm, strict vegan diet, and medication associations such as metformin or long-term acid suppression. If both folate and B12 deficiency are possible, replace B12 first or together; folate alone can improve anemia while neurologic injury from B12 deficiency progresses.

## Folate deficiency

Folate stores are smaller than B12 stores, so deficiency can develop over months. Folate deficiency causes megaloblastic anemia and hypersegmented neutrophils but does not classically cause the posterior column and corticospinal tract findings of B12 deficiency. Causes include poor nutrition, alcohol use, pregnancy, chronic hemolysis, malabsorption, and antifolate medications. Homocysteine rises in both folate and B12 deficiency, while methylmalonic acid rises mainly in B12 deficiency.

## 8. Hemolytic Anemia: The Entry Framework

Hemolysis means premature RBC destruction. If marrow reserve is intact, the reticulocyte count rises. Hemolysis can be intravascular, where RBCs rupture within circulation, or extravascular, where splenic and hepatic macrophages remove abnormal or antibody-coated RBCs. It can be inherited or acquired, immune or nonimmune, membrane-based, enzyme-based, hemoglobin-based, mechanical, infectious, toxic, or microangiopathic.

| Lab/smear clue                    | Interpretation                                                                                                                          |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| High reticulocyte index           | Marrow response to RBC loss/destruction.                                                                                                |
| High LDH                          | Cell breakdown; often prominent in intravascular hemolysis and ineffective erythropoiesis.                                              |
| Low haptoglobin                   | Free plasma hemoglobin binds haptoglobin; low values support intravascular hemolysis but can be affected by liver disease/inflammation. |
| High indirect bilirubin           | Heme breakdown overwhelms hepatic conjugation capacity.                                                                                 |
| Hemoglobinuria/hemosiderinuria    | Intravascular hemolysis with filtered hemoglobin or iron in urine.                                                                      |
| Schistocytes                      | Mechanical fragmentation: TTP/HUS, DIC, malignant hypertension, prosthetic valves, HELLP.                                               |
| Spherocytes                       | Membrane loss from hereditary spherocytosis or warm autoimmune hemolysis.                                                               |
| Positive direct antiglobulin test | Immune hemolysis from IgG and/or complement on RBC surface.                                                                             |

Because the uploaded segment only begins the hemolysis chapter, this section provides the diagnostic scaffold rather than full treatment details. A later hemolysis/hemoglobinopathy chapter should expand autoimmune hemolytic anemia, hereditary spherocytosis, G6PD deficiency, pyruvate kinase deficiency, PNH, sickle cell disease, MAHA, TTP/HUS, DIC, and transfusion strategy in hemoglobinopathies.



## 9. High-Yield Exam Clues and Common Pitfalls

| Clue                                                               | Likely diagnosis or principle                | Why it matters                                                                                     |
|--------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------|
| Elderly man with iron deficiency anemia                            | Occult GI blood loss until proven otherwise. | Do not simply prescribe iron and stop; colon cancer must be considered.                            |
| Microcytosis with normal/high RBC count                            | Thalassemia trait.                           | Avoid prolonged unnecessary iron if iron studies are normal.                                       |
| Low iron + low TIBC + high ferritin                                | Anemia of inflammation.                      | Iron is present but unavailable due to hepcidin-driven sequestration.                              |
| Macrocytosis + hypersegmented neutrophils                          | B12 or folate deficiency.                    | Check B12 before folate-only therapy if neurologic risk exists.                                    |
| Macrocytosis + high reticulocytes                                  | Hemolysis or bleeding recovery.              | Reticulocytes are large; macrocytosis is not always B12/folate.                                    |
| Pancytopenia + hypocellular marrow                                 | Aplastic anemia.                             | Infection and bleeding risk dominate management.                                                   |
| Fever/flank pain/hypotension during transfusion                    | Acute hemolytic transfusion reaction.        | Stop transfusion immediately and notify blood bank.                                                |
| Dyspnea + hypertension after transfusion                           | TACO more likely than TRALI.                 | Volume overload responds to diuretics; prevention is slower transfusion/diuresis in risk patients. |
| Dyspnea + hypotension + noncardiogenic pulmonary edema within 6 hr | TRALI.                                       | Supportive respiratory care; not primarily a fluid overload problem.                               |

### Common pitfalls

- Using MCV alone before reticulocyte response. A high-retic anemia can be normocytic or macrocytic depending on timing.
- Calling iron deficiency in every microcytic anemia. Thalassemia trait and anemia of inflammation are common mimics.
- Ignoring ferritin's acute-phase behavior. Inflammation can hide iron deficiency by raising ferritin.
- Giving folate alone when B12 deficiency has not been excluded. The anemia can improve while neurologic injury progresses.
- Assuming a stable chronic anemia needs transfusion solely because hemoglobin is below normal. Treat mechanism and symptoms, not only the lab flag.
- Missing dilutional anemia after IV fluid resuscitation or pregnancy-related plasma volume expansion.
- Forgetting that acute hemorrhage may have a near-normal initial hemoglobin before equilibration.

## 10. Integrated Practice Questions

### Case 1

**Question:** A 68-year-old man has fatigue, Hb 9.4 g/dL, MCV 72 fL, ferritin 8 ng/mL, high TIBC, and high RDW. He denies visible bleeding. What is the key next clinical step after confirming iron deficiency?

**Answer:** Evaluate for occult GI blood loss, including colon cancer risk assessment and endoscopic evaluation as appropriate. Iron replacement treats the deficit but does not identify the cause.

### Case 2

**Question:** A 24-year-old woman has Hb 11.2 g/dL, MCV 64 fL, normal ferritin, normal RDW, and a relatively high RBC count. What diagnosis rises on the differential?

**Answer:** Thalassemia trait. Hemoglobin electrophoresis helps identify beta-thalassemia trait; alpha-thalassemia trait can have normal electrophoresis after infancy and may require genetic testing if clinically important.

### Case 3

**Question:** A hospitalized patient with rheumatoid arthritis has Hb 10.1 g/dL, MCV 84 fL, low serum iron, low TIBC, and ferritin 300 ng/mL. What mechanism explains the anemia?



**Answer:** Inflammation-mediated hepcidin elevation blocks iron export from macrophages and enterocytes, producing functional iron restriction and underproduction.

#### Case 4

**Question:** A patient with macrocytic anemia has paresthesias, impaired vibration sense, hypersegmented neutrophils, high methylmalonic acid, and high homocysteine. What should be replaced?

**Answer:** Vitamin B12. Folate alone is unsafe when B12 deficiency is possible because neurologic injury can progress despite hematologic improvement.

#### Case 5

**Question:** A patient develops fever, hypotension, flank pain, and red urine 20 minutes into a transfusion. What is the immediate action?

**Answer:** Stop the transfusion, maintain IV access with normal saline, verify identification, notify the blood bank, and begin supportive management for suspected acute hemolytic transfusion reaction.

#### Case 6

**Question:** A patient has Hb 8.5 g/dL, elevated LDH, low haptoglobin, indirect hyperbilirubinemia, and schistocytes. What category of anemia is this?

**Answer:** Hemolytic anemia with microangiopathic/mechanical fragmentation. The next diagnostic branch includes TTP/HUS, DIC, malignant hypertension, HELLP, and prosthetic valve hemolysis depending on the clinical setting.

#### Case 7

**Question:** A patient with pancytopenia has a hypocellular bone marrow biopsy. What anemia category does this represent?

**Answer:** Aplastic anemia or marrow failure physiology. The clinical danger is not only anemia but also neutropenic infection and thrombocytopenic bleeding.

#### Case 8

**Question:** A heavy alcohol user has macrocytosis with normal B12 and folate levels. What are two possible mechanisms?

**Answer:** Alcohol can directly suppress marrow and cause nonmegaloblastic macrocytosis; liver disease can alter RBC membrane lipids. Folate deficiency may still coexist and should be assessed clinically.

## 11. Graph-RAG Extraction Map

The following map is designed for future graph extraction. Each node should retain synonyms, diagnostic thresholds, mechanistic relationships, and source-document context. Edges should be named because the relationship type is often more important than co-occurrence.

| Node                   | Class                        | Core attributes                                                                                            | High-value edges                                                                                                     |
|------------------------|------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Anemia                 | Syndrome                     | Low RBC mass/oxygen-carrying capacity; measured by Hb/Hct; acute vs chronic; symptomatic vs asymptomatic.  | causes -> tissue hypoxia; evaluated_by -> CBC, reticulocyte index, smear; treated_by -> mechanism-specific therapy.  |
| Reticulocyte index     | Diagnostic test              | Corrected marrow response; >2 suggests response, <2 suggests underproduction.                              | separates -> hemolysis/blood loss from underproduction; depends_on -> marrow substrate, EPO, marrow integrity.       |
| Iron deficiency anemia | Microcytic anemia            | Low ferritin, low iron, high TIBC, high RDW; adult men/postmenopausal women require blood loss evaluation. | caused_by -> chronic blood loss; treated_by -> iron replacement; mimicked_by -> thalassemia, anemia of inflammation. |
| Anemia of inflammation | Normocytic/microcytic anemia | Low iron, low/normal TIBC, normal/high ferritin; hepcidin-mediated iron sequestration.                     | caused_by -> chronic infection/inflammation/malignancy/CKD; mechanism -> ferroportin degradation.                    |
| Thalassemia            | Inherited microcytic anemia  | Reduced alpha or beta globin synthesis; target cells; electrophoresis useful for beta forms.               | mimics -> iron deficiency; complicated_by -> iron overload if transfusion-dependent.                                 |

| Node                   | Class                   | Core attributes                                                                         | High-value edges                                                                                                        |
|------------------------|-------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Sideroblastic anemia   | Heme synthesis disorder | Ring sideroblasts; high iron/ferritin; causes include alcohol, lead, INH, MDS.          | caused_by -> defective heme synthesis; treated_by -> remove cause, pyridoxine in select cases.                          |
| Vitamin B12 deficiency | Megaloblastic anemia    | Macrocytosis, hypersegmented neutrophils, neurologic injury, high MMA and homocysteine. | caused_by -> pernicious anemia, ileal disease, malabsorption; must_precede -> folate-only therapy.                      |
| Folate deficiency      | Megaloblastic anemia    | Macrocytosis, high homocysteine, normal MMA, no classic posterior column syndrome.      | caused_by -> alcohol, pregnancy, poor diet, antifolate drugs; treated_by -> folic acid.                                 |
| Aplastic anemia        | Marrow failure          | Pancytopenia, low reticulocytes, hypocellular marrow.                                   | causes -> infection risk, bleeding risk; treated_by -> support, immunosuppression, transplant.                          |
| Hemolytic anemia       | High-retic anemia       | LDH high, haptoglobin low, indirect bilirubin high; intravascular vs extravascular.     | subtyped_by -> DAT, smear morphology, site of destruction; causes -> jaundice, gallstones, hemoglobinuria.              |
| Transfusion reaction   | Iatrogenic complication | Acute or delayed; immune, pulmonary, volume, allergic, infectious categories.           | requires -> stop transfusion; evaluated_by -> blood bank workup; prevents_by -> matching, leukoreduction, rate control. |

## Candidate relationship triples

| Subject                              | Predicate                 | Object                                                               |
|--------------------------------------|---------------------------|----------------------------------------------------------------------|
| Iron deficiency anemia               | has_iron_studies          | low ferritin, low serum iron, high TIBC, low transferrin saturation  |
| Anemia of inflammation               | mediated_by               | hepcidin elevation and ferroportin degradation                       |
| Beta-thalassemia trait               | often_has                 | microcytosis disproportionate to anemia with normal RDW              |
| Vitamin B12 deficiency               | causes                    | subacute combined degeneration and megaloblastic anemia              |
| Folate deficiency                    | distinguished_from_B12_by | normal methylmalonic acid and absence of classic neurologic findings |
| Aplastic anemia                      | diagnosed_by              | hypocellular bone marrow biopsy in pancytopenia                      |
| Acute hemolytic transfusion reaction | caused_by                 | ABO-incompatible intravascular hemolysis                             |
| Reticulocyte index greater than 2    | suggests                  | blood loss or hemolysis                                              |
| Reticulocyte index less than 2       | suggests                  | inadequate marrow production                                         |
| Schistocytes                         | suggest                   | microangiopathic or mechanical hemolysis                             |

## 12. RBC Production Physiology: The Marrow as a Supply Chain

A useful way to understand anemia is to treat erythropoiesis as a supply chain. The kidney senses oxygen delivery and releases erythropoietin; marrow progenitors respond to erythropoietin; DNA synthesis requires vitamin B12 and folate; hemoglobin synthesis requires iron, protoporphyrin, and globin chains; mature RBCs then circulate for about 120 days before macrophage clearance. Failure at any point produces a predictable pattern. Kidney disease produces inadequate signal. Aplastic anemia produces inadequate factory capacity. B12 or folate deficiency produces defective DNA assembly. Iron deficiency and thalassemia produce defective hemoglobin packaging. Hemolysis and hemorrhage increase demand and shorten effective RBC lifespan.

The marrow response is not instantaneous. Reticulocytes usually rise several days after acute bleeding or hemolysis if iron, folate, B12, marrow space, and EPO signaling are adequate. A patient evaluated very early after bleeding may not yet have a high reticulocyte response. Conversely, chronic hemolysis can maintain a high reticulocyte count for months or years until folate depletion, marrow exhaustion, infection, renal injury, or inflammatory suppression blunts compensation.

| Supply-chain component      | Failure pattern                                      | Representative diagnoses                                           | Expected clue                                                  |
|-----------------------------|------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------|
| Oxygen sensor/EPO signal    | Underproduction with low reticulocytes               | CKD, inflammatory EPO resistance, endocrine disease                | Normocytic anemia, low retic, kidney disease context           |
| Marrow stem/progenitor pool | Multilineage cytopenias or isolated red cell aplasia | Aplastic anemia, marrow infiltration, myelodysplasia, chemotherapy | Low retic; abnormal WBC/platelets; marrow biopsy may be needed |

| Supply-chain component | Failure pattern                                                      | Representative diagnoses                                          | Expected clue                                                 |
|------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------|
| DNA synthesis          | Megaloblastic underproduction                                        | B12 deficiency, folate deficiency, methotrexate, hydroxyurea      | Macro-ovalocytes, hypersegmented neutrophils, high MCV        |
| Hemoglobin synthesis   | Microcytic underproduction                                           | Iron deficiency, thalassemia, sideroblastic anemia, lead toxicity | Low MCV, hypochromia, abnormal iron studies or globin testing |
| RBC survival           | High-retic anemia if marrow can respond                              | Hemolysis, hypersplenism, mechanical valves, immune hemolysis     | High LDH/indirect bilirubin, low haptoglobin, smear clues     |
| Blood volume integrity | High-retic anemia after delay; acute instability may precede Hb drop | GI bleed, trauma, gynecologic bleeding, postoperative bleeding    | Tachycardia, orthostasis, melena, falling serial Hb           |

## 13. Practical Lab Ordering: Minimal, Focused, and Escalating Workups

A good anemia workup is neither random nor overly narrow. The first pass should answer whether anemia is real, whether it is dangerous, whether the marrow is responding, and which broad morphologic pathway is likely. A second pass should confirm mechanism and source. A third pass should pursue rare or marrow-level diagnoses when common explanations fail or when multiple cell lines are abnormal.

| Clinical situation        | Initial tests                                                                                                 | Second-line tests                                                                                                   | Escalation triggers                                                                                                |
|---------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| New stable anemia         | Repeat CBC if unexpected, reticulocyte count/index, MCV/RDW, smear, iron studies, B12/folate, creatinine/eGFR | TSH, LFTs, CRP/ESR, hemolysis labs if retic high, stool/gynecologic evaluation if iron deficient                    | Unexplained cytopenias, abnormal smear, severe anemia, constitutional symptoms, splenomegaly                       |
| Suspected iron deficiency | Ferritin, iron, TIBC/transferrin, transferrin saturation, CBC/RDW                                             | Celiac testing or H pylori context when appropriate; GI/gynecologic blood loss evaluation                           | Men, postmenopausal women, weight loss, positive stool blood, recurrent or refractory iron deficiency              |
| Suspected hemolysis       | Reticulocytes, LDH, haptoglobin, indirect bilirubin, DAT, smear                                               | G6PD level after acute episode resolves, PNH flow cytometry, hemoglobin electrophoresis, ADAMTS13 when TTP possible | Schistocytes, thrombocytopenia, renal injury, neurologic signs, severe jaundice, hemoglobinuria                    |
| Macrocytosis              | B12, folate, smear, retic, medication/alcohol review                                                          | MMA, homocysteine, TSH, LFTs                                                                                        | Cytopenias, dysplasia, unexplained persistent macrocytosis, older patient                                          |
| Pancytopenia              | CBC with differential, retic, smear, B12/folate, HIV/hepatitis when appropriate                               | Hemolysis labs, inflammatory/infectious testing, spleen imaging if indicated                                        | Blasts, severe neutropenia, severe thrombocytopenia, hypocellular or infiltrative concern - bone marrow evaluation |

The most efficient rule is to let the reticulocyte index dictate tempo. High reticulocytes require urgent exclusion of bleeding and hemolysis. Low reticulocytes require substrate, kidney, inflammatory, endocrine, and marrow assessment. A smear should be reviewed whenever the anemia is severe, unexplained, associated with other cytopenias, associated with hemolysis markers, or discordant with automated indices.

## 14. Microcytic Anemia Illness Scripts

**Iron deficiency script:** chronic fatigue or exertional dyspnea plus microcytosis, high RDW, low ferritin, and low transferrin saturation. Symptoms can include pica, pagophagia, restless legs, brittle nails, glossitis, and poor exercise tolerance. In adults without an obvious menstrual or pregnancy explanation, the disease label is incomplete until bleeding source, malabsorption, diet, and medication contributors have been considered.

**Anemia of inflammation script:** patient with chronic inflammatory disease, infection, malignancy, CKD, or severe systemic illness has mild-to-moderate anemia, often normocytic at first, with low serum iron but ferritin that is normal or high. The marrow is not truly iron-empty; iron is locked away by hepcidin. This is why oral iron alone may have limited effect unless absolute iron deficiency coexists.

**Thalassemia trait script:** young or asymptomatic patient from an at-risk ancestry background has marked microcytosis with mild anemia, normal iron stores, target cells, and an RBC count that is preserved relative to the hemoglobin. The RDW may be normal because cells are uniformly small. The diagnosis matters because iron therapy is not the solution unless true deficiency is also present, and reproductive counseling may be important.

Sideroblastic script: patient has microcytic or dimorphic anemia with high iron availability, possible basophilic stippling, and risk factors such as alcohol use, isoniazid, linezolid, chloramphenicol, lead exposure, copper deficiency, or myelodysplasia. The marrow stores iron but cannot build heme effectively. Treatment is cause removal, supportive care, and pyridoxine responsiveness in select forms.

| Microcytic diagnosis   | One-sentence mechanism                              | Do not miss                                                        |
|------------------------|-----------------------------------------------------|--------------------------------------------------------------------|
| Iron deficiency        | Not enough usable iron to build hemoglobin.         | Adult GI malignancy or chronic blood loss source.                  |
| Anemia of inflammation | Hepcidin traps iron and suppresses marrow response. | Coexisting true iron deficiency can hide inside inflammation.      |
| Thalassemia            | Globin-chain production is genetically reduced.     | Normal iron studies with very low MCV and target cells.            |
| Sideroblastic anemia   | Iron reaches mitochondria but heme synthesis fails. | High iron/ferritin despite microcytosis; MDS or toxin/drug causes. |

## 15. Macrocytic and Multilineage Illness Scripts

Megaloblastic anemia script: macrocytosis, hypersegmented neutrophils, low reticulocyte production, and ineffective erythropoiesis. Because marrow precursors fail to divide normally, the marrow may be cellular but ineffective, and intramedullary cell death can raise LDH and indirect bilirubin, mimicking hemolysis. The presence of neurologic findings pushes strongly toward B12 deficiency rather than isolated folate deficiency.

B12 deficiency neurologic script: paresthesias, impaired vibration/position sense, ataxia, cognitive changes, spasticity, and urinary symptoms may occur with or without severe anemia. Posterior column and corticospinal tract involvement creates the classic subacute combined degeneration pattern. This is one of the most important reversible neurologic diagnoses in medicine, but reversibility declines when treatment is delayed.

Aplastic anemia script: fatigue, infections, mucosal bleeding, petechiae, and bruising occur together because all three major hematopoietic lines are reduced. The marrow is empty or hypocellular rather than packed. The differential includes drugs, toxins, viral triggers, immune-mediated disease, inherited syndromes, and hypocellular myelodysplasia. Management depends on age, severity, donor availability, and comorbidities.

Myelodysplasia caution: persistent macrocytosis with cytopenias in an older adult should not be assumed to be alcohol or nutrition without confirmation. Dysplastic neutrophils, abnormal platelets, unexplained cytopenias, and failure to respond to vitamin replacement should prompt hematology evaluation and bone marrow consideration.

## 16. Management Principles by Mechanism

Mechanism-specific treatment is safer than reflex treatment. Iron deficiency requires iron repletion and source evaluation. Anemia of inflammation requires control of the underlying disease and selective iron/EPO strategies when appropriate. B12 deficiency requires B12 replacement and cause identification. Folate deficiency requires folate replacement after B12 deficiency is excluded or simultaneously treated. Hemolysis requires subtype diagnosis because steroids, transfusion, complement inhibition, splenectomy, avoidance of oxidants, plasma exchange, anticoagulation, or supportive care may be appropriate depending on the cause.

| Mechanism               | Immediate management question                                                                         | Definitive management question                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Acute blood loss        | Is the patient unstable and does bleeding need resuscitation/source control now?                      | What is the bleeding source and how is recurrence prevented?                              |
| Iron deficiency         | Is the anemia severe/symptomatic enough for urgent transfusion or IV iron?                            | Why is iron missing: GI, gynecologic, dietary, malabsorptive, or medication-related?      |
| B12 deficiency          | Are neurologic findings present and should parenteral therapy be used?                                | Is this pernicious anemia, ileal disease, diet, medication, or malabsorption?             |
| Aplastic/marrow failure | Are neutropenia or thrombocytopenia creating immediate infection/bleeding danger?                     | Is transplant, immunosuppression, or disease-specific therapy indicated?                  |
| Hemolysis               | Is this life-threatening MAHA, immune hemolysis, severe sickle complication, or transfusion reaction? | Which subtype explains the smear, DAT, inherited risk, medications, and clinical setting? |

Transfusion is supportive, not curative. In iron deficiency, transfusion may temporarily improve oxygen carrying capacity but does not replete total body iron enough to fix the problem. In thalassemia major, transfusion sustains life but creates iron overload risk. In autoimmune hemolysis, transfusion may be necessary but crossmatching can be difficult. In TTP-like MAHA, platelet transfusion may be harmful unless life-threatening bleeding exists. The product must match the mechanism.

## 17. Expanded Differential Tables for RAG Retrieval

| Clinical phrase                       | Retrieve these concepts                                                                | Reasoning link                                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Fatigue plus pica                     | Iron deficiency, ferritin, GI blood loss, pregnancy, menstrual bleeding                | Pica/pagophagia are classic behavioral clues to iron deficiency.                                    |
| Microcytosis with normal ferritin     | Thalassemia, anemia of inflammation, sideroblastic anemia                              | Iron deficiency becomes less likely when iron stores are normal unless inflammation masks ferritin. |
| Macrocytosis plus neuropathy          | B12 deficiency, pernicious anemia, subacute combined degeneration                      | Neurologic deficits separate B12 from folate in classic teaching.                                   |
| Anemia plus renal disease             | EPO deficiency, anemia of CKD, iron restriction, ESA therapy                           | Kidney failure lowers EPO signaling and often adds inflammatory iron restriction.                   |
| Anemia plus low platelets and low WBC | Aplastic anemia, marrow infiltration, B12/folate severe deficiency, MDS, hypersplenism | Multilineage cytopenia implies marrow/system-level pathology.                                       |
| Jaundice plus high reticulocytes      | Hemolysis, indirect bilirubin, LDH, haptoglobin, DAT                                   | RBC destruction increases bilirubin and marrow compensation.                                        |
| Schistocytes plus thrombocytopenia    | TTP, HUS, DIC, malignant hypertension, HELLP                                           | Fragmentation hemolysis suggests small-vessel or mechanical RBC injury.                             |
| Fever during transfusion              | Acute hemolytic reaction, febrile nonhemolytic reaction, sepsis, TRALI                 | Stop first, sort mechanism second.                                                                  |

## 18. Additional Board-Style Mini Cases

### Mini-case 1

**Stem:** A patient with long-standing Crohn disease and ileal resection develops gait instability, numb feet, MCV 112 fL, and hypersegmented neutrophils.

**Teaching point:** Think vitamin B12 deficiency from terminal ileal malabsorption. Check B12 with methylmalonic acid if borderline, replace B12, and do not rely on folate alone.

### Mini-case 2

**Stem:** A patient taking isoniazid develops anemia with high serum iron and ring sideroblasts in marrow.

**Teaching point:** Think sideroblastic anemia from impaired heme synthesis. Remove or address the cause and consider pyridoxine responsiveness.

### Mini-case 3

**Stem:** A patient with CKD stage 4 has normocytic anemia, low retic index, and no bleeding.

**Teaching point:** Think inadequate EPO production plus inflammatory iron restriction. Check iron status before considering ESA-based management.

### Mini-case 4

**Stem:** A patient with rheumatoid arthritis has low serum iron but high ferritin.

**Teaching point:** Think anemia of inflammation rather than simple iron deficiency. Hepcidin is the central mechanism.

### Mini-case 5

**Stem:** A young asymptomatic patient has MCV 62 fL, Hb 11.8 g/dL, normal ferritin, normal RDW, and target cells.

**Teaching point:** Think thalassemia trait. Avoid reflex iron unless deficiency is proven.

### Mini-case 6

**Stem:** An older adult has macrocytosis, neutropenia, thrombocytopenia, and no response to B12/folate replacement.

**Teaching point:** Think myelodysplastic syndrome or marrow disorder. Smear and bone marrow evaluation become important.

### Mini-case 7

**Stem:** During transfusion, a patient develops respiratory distress, hypertension, JVD, and pulmonary edema.

**Teaching point:** Think TACO. Stop transfusion, give oxygen and diuresis; future transfusions should be slower with volume strategy.

### Mini-case 8

**Stem:** During transfusion, a patient develops acute hypoxemia and bilateral infiltrates without signs of volume overload.

**Teaching point:** Think TRALI. Provide supportive respiratory care and notify transfusion service.

### Mini-case 9

**Stem:** A patient has severe anemia and a normal MCV, but ferritin is very low.

**Teaching point:** Early iron deficiency or mixed anemia can be normocytic. MCV does not exclude iron deficiency.

### Mini-case 10

**Stem:** A patient has anemia, indirect hyperbilirubinemia, very high LDH, low haptoglobin, and a negative stool blood test.

**Teaching point:** Think hemolysis. Use smear and DAT to separate immune, membrane, enzyme, hemoglobin, and fragmentation causes.

## 19. Source and Accuracy Notes

This chapter uses the uploaded hematology/anemia segment as a coverage signal rather than a source to copy. It expands the covered topics with original explanatory text, original tables, original algorithms, and original graph-RAG maps. Clinical practice varies by institution; transfusion thresholds, iron repletion route, endoscopic evaluation, and marrow workup should be individualized to patient context and local protocols.

Key external clinical anchors used for cross-checking include modern restrictive transfusion guidance, contemporary iron deficiency anemia guidance, and hematology society resources for hemoglobinopathy/transfusion principles. These sources support the broad framing that stable adult transfusion is often restrictive around hemoglobin 7 g/dL, that iron deficiency in adults requires diagnosis plus source evaluation, and that hemoglobinopathy transfusion requires specialized matching and complication prevention.